

Seminar

General Spatio-Temporal Factor Models for High-Dimensional Random Fields on a Lattice (Barigozzi, La Vecchia & Liu)

Methodology description (Replication Report)

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1 Introduction

High-dimensional spatio-temporal datasets — such as temperature or energy observations measured on spatial grids over time — pose severe challenges due to the rapid growth in the number of parameters. Classical models such as VAR, SAR, or spatio-temporal autoregressive specifications can become unstable or infeasible when the number of spatial locations and time points is large, motivating the need for dimension reduction.

General Dynamic Factor Models (GDFM) provide an effective framework for high-dimensional time series, but they do not explicitly account for spatial dependence. A naive solution is to stack spatial series into a single large vector and apply a standard factor model; however, this approach ignores the lattice structure of the data and may result in spurious factors or weak identification.

Barigozzi et al. propose **General Spatio-Temporal Factor Models (GSTFM)** for high-dimensional random fields observed on a lattice. Their framework decomposes the observed field into a **common component** and an **idiosyncratic component**, where the common component is driven by a small number q of latent factors. Estimating q is therefore crucial to recover the dominant spatio-temporal dynamics.

This report explains why standard GDFM techniques may fail in lattice-structured settings, summarizes the theoretical solution provided by q-GSTFM, and evaluates the method empirically using **WeatherBench2** data.

2 Motivation and setup

2.1 Setup

We observe a high-dimensional spatio-temporal random field on a lattice.

Let the spatio-temporal index be

$$\varsigma = (s_1, s_2, t)^\top, \quad s_1 = 1, \dots, S_1, \quad s_2 = 1, \dots, S_2, \quad t = 1, \dots, T.$$

For each cross-sectional unit $\ell = 1, \dots, n$, we observe the scalar field $\{x_{\ell\varsigma}\}$ and collect all units in the n -dimensional vector

$$x_{n\varsigma} = (x_{1\varsigma}, \dots, x_{n\varsigma})^\top.$$

2.2 The General Spatio-Temporal Factor Model (GSTFM). Model A.

At each space-time point $\varsigma = (s_1, s_2, t)$, we observe an n -dimensional vector:

$$x_{\ell,\varsigma} = \chi_{\ell,\varsigma} + \xi_{\ell,\varsigma},$$

where $\chi_{\ell,\varsigma}$ is the common component driven by q latent spatio-temporal factors, and $\xi_{\ell,\varsigma}$ is a weakly dependent idiosyncratic term.

GSTFM allows factors to influence both neighboring locations and lagged times. The core idea is that the spectral density matrix of x exhibits exactly q diverging eigenvalues as the dimension $n \rightarrow \infty$, while the remaining eigenvalues stay bounded. This separation — the **eigen-gap** — identifies the number of common factors and represents a “blessing of dimensionality.”

Under a spatio-temporal factor structure, the largest q eigenvalues increase with the cross-sectional dimension m , while the $(q + 1)$ -th eigenvalue remains relatively stable.

Formally, letting $\hat{\lambda}_{x,j}^{(m)}(\theta)$ denote the j -th largest eigenvalue of the estimated spectral density matrix at frequency θ , we consider the frequency-averaged eigenvalues

$$\bar{\lambda}_j(m) = \frac{1}{|\Theta|} \sum_{\theta \in \Theta} \hat{\lambda}_{x,j}^{(m)}(\theta), \quad j = 1, \dots, q + 1.$$

The widening separation between $\bar{\lambda}_q(m)$ and $\bar{\lambda}_{q+1}(m)$ provides empirical evidence of the eigen-gap and supports identification of q . The simulation outcome is shown in Figure~1.

2.3 General Dynamic Factor Models (GDFM). Model B.

GDFM handle high-dimensional time series but ignore spatial dependence. Simply stacking spatial series into one large vector destroys the lattice structure and can lead to spurious factors or loss of identification.

Stacking all space–time series into one vector $x_{N,t}$ of dimension $N = n \times S_1 \times S_2$ and estimating a standard GDFM erases spatial structure. The resulting eigenvalues grow together with no clear gap, making the number of factors unidentifiable.

To illustrate why standard GDFM approaches can be unreliable for lattice-structured data, we stack the spatio-temporal field into a high-dimensional vector time series

$$Z \in \mathbb{R}^{T \times N}, \quad N = nS_1S_2,$$

and apply a conventional spectral density estimator to the stacked process.

When the spatial structure is ignored, the leading eigenvalues tend to grow together as m increases, so the eigen-gap weakens or disappears. This implies that the factor number becomes difficult to identify, even if a spatio-temporal common component is present. Figure~2 illustrates this phenomenon and motivates the use of GSTFM, which preserves the space-time dependence encoded by the lattice.

3 Model and theoretical background

3.1 Notation and spatio-temporal random field on a lattice

We observe a high-dimensional spatio-temporal random field on a lattice. Let the spatio-temporal index be

$$\varsigma = (s_1, s_2, t)^\top, \quad s_1 = 1, \dots, S_1, \quad s_2 = 1, \dots, S_2, \quad t = 1, \dots, T.$$

For each cross-sectional unit $\ell = 1, \dots, n$, we observe the scalar field $\{x_{\ell\varsigma}\}$ and collect all units in the n -dimensional vector

$$x_{n\varsigma} = (x_{1\varsigma}, \dots, x_{n\varsigma})^\top.$$

Throughout, $\theta = (\theta_1, \theta_2, \theta_3)^\top \in [-\pi, \pi]^3$ denotes a spatio-temporal frequency and $h = (h_1, h_2, h_3)^\top \in \mathbb{Z}^3$ a spatio-temporal lag.

3.2 Population model: the q-GSTFM decomposition

The General Spatio-Temporal Factor Model (GSTFM) decomposes each observation into a *common* and an *idiosyncratic* component:

$$x_{\ell\varsigma} = \chi_{\ell\varsigma} + \xi_{\ell\varsigma}, \quad \ell = 1, \dots, n. \quad (1)$$

The common component is driven by a small number $q \ll n$ of latent spatio-temporal factors $u_\varsigma = (u_{1\varsigma}, \dots, u_{q\varsigma})^\top$ through a (possibly infinite) linear filter:

$$\chi_{\ell\varsigma} = b_\ell(L) u_\varsigma = \sum_{\kappa \in \mathbb{Z}^3} \sum_{j=1}^q b_{\ell j, \kappa} u_{j, \varsigma - \kappa}. \quad (2)$$

The term $\xi_{\ell\varsigma}$ captures local effects and is assumed to be weakly cross-sectionally correlated, so that it does not generate diverging eigenvalues as $n \rightarrow \infty$.

3.3 Spectral density matrix and the eigen-gap condition

Define the spatio-temporal autocovariance matrix at lag h as

$$\Gamma_n^x(h) = \mathbb{E} \left[x_{n, \varsigma+h} x_{n\varsigma}^\top \right].$$

The associated $n \times n$ spatio-temporal spectral density matrix is

$$\Sigma_n^x(\theta) = \frac{1}{8\pi^3} \sum_{h \in \mathbb{Z}^3} \Gamma_n^x(h) e^{-i\langle h, \theta \rangle}. \quad (3)$$

Let $\lambda_{n1}^x(\theta) \geq \dots \geq \lambda_{nn}^x(\theta)$ be the eigenvalues of $\Sigma_n^x(\theta)$. The key feature that enables identification of the GSTFM is the *eigen-gap* in the dynamic eigenvalues:

for (Lebesgue-)almost every $\theta \in [-\pi, \pi)^3$,

$$\lambda_{nq}^x(\theta) \rightarrow \infty \quad \text{while} \quad \sup_n \lambda_{n,q+1}^x(\theta) < \infty.$$

Intuitively, the first q eigenvalues correspond to the pervasive common dynamics, whereas the remaining eigenvalues correspond to idiosyncratic variation.

3.4 Population projection and the common component

Let $P_{n,q}^x(\theta)$ denote the matrix collecting the q eigenvectors corresponding to the largest q eigenvalues of $\Sigma_n^x(\theta)$. Define the rank- q projection matrix

$$K_n^x(\theta) = P_{n,q}^x(\theta) P_{n,q}^x(\theta)^\dagger.$$

The corresponding linear filter coefficients (in the lag domain) are obtained via inverse Fourier transform:

$$K_n^x(h) = \frac{1}{8\pi^3} \int_{[-\pi, \pi)^3} K_n^x(\theta) e^{i\langle h, \theta \rangle} d\theta.$$

Then the population common component can be expressed as a filtered version of the observed field:

$$\chi_{\ell_\varsigma} = \sum_{m=1}^n \sum_{h \in \mathbb{Z}^3} K_{n,\ell_m}^x(h) x_{m,\varsigma-h}. \quad (4)$$

4 Methodology

4.1 Estimation: kernel-smoothed spectral density matrix

In finite samples, $\Sigma_n^x(\theta)$ is unknown and must be estimated. The method uses a kernel-smoothed estimator of the spectral density matrix. Let K_1, K_2, K_3 be kernel functions and let (B_{S_1}, B_{S_2}, B_T) be bandwidth parameters. The estimator is

$$\hat{\Sigma}_n^x(\theta) = \frac{1}{S_1 S_2 T} \sum_{\varsigma_1} \sum_{\varsigma_2} x_{n, \varsigma_1} x_{n, \varsigma_2}^\top K_1\left(\frac{s_{11} - s_{21}}{B_{S_1}}\right) K_2\left(\frac{s_{12} - s_{22}}{B_{S_2}}\right) K_3\left(\frac{t_1 - t_2}{B_T}\right) e^{-i\langle \varsigma_1 - \varsigma_2, \theta \rangle}. \quad (5)$$

In practice, (5) is evaluated on a discrete frequency grid $\{\theta_h\}_{h=1}^{S_1 S_2 T}$ corresponding to spatio-temporal Fourier frequencies.

4.2 Estimation of the common component

For each frequency θ_h , compute the eigen-decomposition of $\hat{\Sigma}_n^x(\theta_h)$ and retain the $\hat{q}^{(n)}$ leading eigenvectors, stacked in $\hat{P}_{n, \hat{q}}^x(\theta_h)$. The estimated projection matrix is

$$\hat{K}_n^x(\theta_h) = \hat{P}_{n, \hat{q}}^x(\theta_h) \hat{P}_{n, \hat{q}}^x(\theta_h)^\dagger.$$

The estimated filter coefficients are obtained via inverse Fourier transform. Since the theoretical filter is two-sided and infinite, it is approximated by truncation using integers (M_{S_1}, M_{S_2}, M_T) :

$$|h_1| \leq M_{S_1}, \quad |h_2| \leq M_{S_2}, \quad |h_3| \leq M_T.$$

Finally, the estimated common component is

$$\hat{\chi}_{\ell_\varsigma}^{(n)} = \sum_{m=1}^n \sum_{|h_1| \leq M_{S_1}} \sum_{|h_2| \leq M_{S_2}} \sum_{|h_3| \leq M_T} \hat{K}_{n, \ell_m}^x(h) x_{m, \varsigma - h}. \quad (6)$$

4.2.1 Algorithm 1: Estimation of the common component

Algorithm 1 Algorithm 1: Estimation of the common component

Require: Data $\{x_{\ell\varsigma}\}$, $\ell = 1, \dots, n$, $\varsigma = (s_1, s_2, t)^\top$; estimated number of factors $\hat{q}^{(n)}$; kernels K_1, K_2, K_3 ; bandwidths (B_{S_1}, B_{S_2}, B_T) ; truncation parameters (M_{S_1}, M_{S_2}, M_T) .

Ensure: Estimated common component $\{\hat{\chi}_{\ell\varsigma}^{(n)}\}$.

- 1: Compute $\hat{\Sigma}_n^x(\theta_h)$ on the frequency grid using (5).
 - 2: **for** each frequency θ_h **do**
 - 3: Compute the leading $\hat{q}^{(n)}$ eigenvectors of $\hat{\Sigma}_n^x(\theta_h)$.
 - 4: Construct $\hat{K}_n^x(\theta_h) = \hat{P}_{n,\hat{q}}^x(\theta_h) \hat{P}_{n,\hat{q}}^x(\theta_h)^\dagger$.
 - 5: **end for**
 - 6: Obtain truncated filter coefficients $\hat{K}_n^x(h)$ by inverse Fourier transform.
 - 7: Compute $\hat{\chi}_{\ell\varsigma}^{(n)}$ using (6).
 - 8: **return** $\{\hat{\chi}_{\ell\varsigma}^{(n)}\}$.
-

4.3 Determining the number of factors

An essential implementation step is to estimate the number of factors q . The method uses an information criterion based on the eigenvalues of $\hat{\Sigma}_n^x(\theta)$. Let $\hat{\lambda}_{nj}^x(\theta)$ denote the j -th largest eigenvalue of $\hat{\Sigma}_n^x(\theta)$. For $k = 0, \dots, q_{\max}$ define

$$IC^{(n)}(k) = \log \left[\frac{1}{n} \sum_{j=k+1}^n \left(\frac{1}{8\pi^3} \int_{[-\pi, \pi]^3} \hat{\lambda}_{nj}^x(\theta) d\theta \right) \right] + k p(n, S_1, S_2, T), \quad (7)$$

where $p(n, S_1, S_2, T)$ is a penalty term that vanishes asymptotically but not too fast. The estimated factor number is then

$$\hat{q}^{(n)} = \arg \min_{0 \leq k \leq q_{\max}} IC^{(n)}(k).$$

4.3.1 Algorithm 2: Selection of the number of factors

Algorithm 2 Algorithm 2: Selection of the number of factors $\hat{q}^{(n)}$

Require: Data $\{x_{\ell\varsigma}\}$; maximum number of factors $q_{\max}$; subsample sizes $n_1 < \dots < n_J = n$; constants $c_1, \dots, c_L$; kernels K_1, K_2, K_3 ; bandwidths (B_{S_1}, B_{S_2}, B_T) .

Ensure: $\hat{q}^{(n)}$.

- 1: Choose a random permutation of the n cross-sectional units.
 - 2: **for** $\ell = 1, \dots, L$ **do**
 - 3: **for** $j = 1, \dots, J$ **do**
 - 4: Compute $\hat{\Sigma}_{n_j}^x(\theta_h)$ on the grid using (5).
 - 5: Compute the $q_{\max}$ largest eigenvalues of $\hat{\Sigma}_{n_j}^x(\theta_h)$.
 - 6: Evaluate $IC^{(n_j)}(k)$ in (7) for $k = 0, \dots, q_{\max}$ using penalty constant c_ℓ .
 - 7: Set $\hat{q}_{c_\ell}^{(n_j)} = \arg \min_{0 \leq k \leq q_{\max}} IC^{(n_j)}(k)$.
 - 8: **end for**
 - 9: Compute a stability/variability score S_{c_ℓ} from $\{\hat{q}_{c_\ell}^{(n_j)}\}_{j=1}^J$.
 - 10: **end for**
 - 11: Choose $\hat{c}$ from the second stability region in the plot $c \mapsto c S_c$.
 - 12: **return** $\hat{q}^{(n)} = \hat{q}_{\hat{c}}^{(n_J)}$.
-

5 Real data application: WeatherBench2 / ERA5

5.1 Data construction

The empirical application uses ERA5 reanalysis weather data provided through the **WeatherBench2** repository. We extract a spatio-temporal block

$$x \in \mathbb{R}^{n \times S_1 \times S_2 \times T},$$

where T denotes the number of time steps, $S_1 \times S_2$ the spatial grid resolution (latitude $\times$ longitude), and n the cross-sectional dimension corresponding to multiple meteorological variables and vertical pressure levels.

5.1.1 Variables

We focus on four standard pressure-level ERA5 fields:

`temperature`, `geopotential`, `u_component_of_wind`, `v_component_of_wind`.

These variables are available on the common four-dimensional grid (time, level, latitude, longitude), ensuring consistent alignment required by the GSTFM procedure.

To keep the computational burden manageable during development and replication, we restrict attention to the time interval January 2010, sampled at the native 6-hourly frequency.

We select a European subdomain defined by the bounding box of latitude and longitude.

This produces a moderately sized spatial lattice while preserving realistic meteorological variability.

The original dataset contains multiple atmospheric pressure levels. For computational efficiency, we retain the first eight levels in the dataset:

$$\{50, 100, 150, 200, 250, 300, 400, 500\} \text{ hPa.}$$

This choice captures key vertical structure in the troposphere while keeping the dimension of the spectral density estimation tractable.

Overall, the extracted block combines the selected variables across the chosen pressure levels and the spatial-temporal subset, providing a realistic high-dimensional spatio-temporal field for the GSTFM estimator.

After subsetting ERA5 over the chosen time window, spatial domain, and the first eight pressure levels, each meteorological variable yields a four-dimensional block of size

$$(8, 23, 27, 124),$$

corresponding to (level, latitude, longitude, time). Since all selected variables share identical spatial and temporal dimensions, we stack the four variables along

the cross-sectional dimension. This produces the raw tensor

$$x_{\text{raw}} \in \mathbb{R}^{32 \times 23 \times 27 \times 124},$$

where $n = 32$ arises from 4 variables $\times$ 8 pressure levels. The resulting dataset therefore represents a high-dimensional spatio-temporal field observed on a 23×27 spatial lattice over $T = 124$ time points.

5.1.2 Preprocessing and standardization

Before estimating the GSTFM, each component series is standardized to remove differences in physical units and variability across variables and pressure levels. Let $x_{\ell, s_1, s_2, t}$ denote the raw extracted field, where $\ell = 1, \dots, n$ indexes the stacked variable-level combinations. For each ℓ , we compute the mean and standard deviation over all spatial locations and times, and define the standardized field

$$x_{\ell, s_1, s_2, t}^{(sc)} = \frac{x_{\ell, s_1, s_2, t} - \mu_\ell}{\sigma_\ell}.$$

This ensures that each series has approximately zero mean and unit variance over the full space-time domain, preventing variables with larger physical scales (e.g., geopotential) from dominating the eigenvalue-based diagnostics.

5.1.3 Bandwidth and truncation choices

The kernel-smoothed estimator of the spatio-temporal spectral density requires bandwidth parameters controlling the amount of smoothing across spatial and temporal lags. Let $x^{(sc)} \in \mathbb{R}^{n \times S_1 \times S_2 \times T}$ denote the standardized field. For the extracted ERA5 subset we have $S_1 = 23$, $S_2 = 27$, and $T = 124$.

In the implementation, we set small bandwidth values

$$B_{S_1} = 2, \quad B_{S_2} = 2, \quad B_T = 2,$$

which provide mild smoothing while keeping computation tractable for this replication.

To reconstruct the common component, the estimated projection filter is theoretically two-sided and infinite. Following the paper, we approximate it by truncating spatial and temporal lags:

$$|h_1| \leq M_{S_1}, \quad |h_2| \leq M_{S_2}, \quad |h_3| \leq M_T.$$

We adopt the conservative truncation choice

$$M_{S_1} = M_{S_2} = M_T = 1,$$

so that the reconstruction uses only local interactions across immediate spatial neighbors and adjacent time steps. These settings represent safe defaults that

ensure numerical stability and fast execution on the subset.

Sensitivity checks with slightly larger bandwidths and truncation windows produce qualitatively similar results.

5.2 Estimated number of factors $\hat{q}$

To determine the number of common components, we computed the eigenvalues of the kernel-smoothed spatio-temporal spectral density estimator $\hat{\Sigma}_n^x(\theta)$ across a discrete grid of frequencies $\theta \in \Theta$. Let $\hat{\lambda}_j(\theta)$ denote the j -th largest eigenvalue of $\hat{\Sigma}_n^x(\theta)$. We summarize the spectral behavior by averaging over the frequency grid:

$$\bar{\lambda}_j = \frac{1}{|\Theta|} \sum_{\theta \in \Theta} \hat{\lambda}_j(\theta), \quad j = 1, \dots, n.$$

The resulting sequence $\{\bar{\lambda}_j\}$ is plotted against the index j . A factor structure is indicated by a sharp drop between the q -th and $(q+1)$ -th eigenvalues. To formalize this diagnostic, we compute the eigenvalue differences (drops)

$$\Delta_j = \bar{\lambda}_j - \bar{\lambda}_{j+1}, \quad j = 1, \dots, q_{\max},$$

and select

$$\hat{q} = \arg \max_{1 \leq j \leq q_{\max}} \Delta_j, \quad \text{with } q_{\max} = 10.$$

As seen on the Figure 3, the largest drop, occurs at $j = 1$, yielding the estimate

$$\hat{q} = 1.$$

This indicates that a single dominant spatio-temporal common component explains most of the cross-sectional dependence in the selected dataset.

5.3 Explained variance of the common component

After applying the GSTFM common component estimator, we obtain $\hat{\chi}$ and $\hat{\xi} = x - \hat{\chi}$.

We summarize the strength of the common component by the explained variance ratio

$$EV = \frac{\|\hat{\chi}\|^2}{\|x\|^2} = \frac{\sum_{\ell, s_1, s_2, t} \hat{\chi}_{\ell, s_1, s_2, t}^2}{\sum_{\ell, s_1, s_2, t} x_{\ell, s_1, s_2, t}^2}.$$

For the weather subset used in this report, we obtain:

$$EV = 0.8522365,$$

meaning the estimated common component explains approximately 85.22% of the variance in the standardized field.

Figure 4 provides a visual check of the GSTFM decomposition at a representative location. The estimated common component $\hat{\chi}$ tracks the main fluctuations

of the standardized series $x^{(sc)}$, indicating that the dominant variation is captured by the shared spatio-temporal factor structure. The idiosyncratic component $\hat{\xi}$ is comparatively smaller in magnitude and absorbs residual, location-specific deviations. This behavior is consistent with the GSTFM interpretation of $\hat{\chi}$ as the pervasive common signal and $\hat{\xi}$ as weakly cross-sectionally dependent noise.

Figure 5 provides a spatial diagnostic of the GSTFM decomposition at a fixed time index $t = 10$ for component $\ell = 1$. The top panel shows the standardized field $x^{(sc)}$, which exhibits a clear large-scale spatial gradient across the domain. The middle panel indicates that the estimated common component $\hat{\chi}$ captures this smooth, coherent pattern, consistent with the idea that the common component represents pervasive spatio-temporal variation shared across the high-dimensional field. The bottom panel displays the idiosyncratic component $\hat{\xi}$, which primarily contains localized deviations and smaller-scale spatial features not explained by the common structure. Overall, the figure supports the interpretation that GSTFM separates broad, structured spatial dependence from residual local variability.

6 Conclusion

This replication study supports the main empirical implication of the General Spatio-Temporal Factor Model (GSTFM): when the data contain a pervasive spatio-temporal factor structure, the largest eigenvalues of the kernel-smoothed spatio-temporal spectral density matrix exhibit a pronounced eigen-gap, enabling identification of the factor dimension q . In contrast, when the spatial structure is removed by stacking the observations into a standard high-dimensional time-series format (as in the GDFM setting), the eigenvalues tend to grow together and the eigen-gap becomes weak or disappears, making factor selection substantially less reliable.

Applying GSTFM to a real WeatherBench2/ERA5 subset over Europe confirms the practical relevance of these diagnostics. The frequency-averaged eigenvalue profile shows a sharp drop after the first eigenvalue, leading to the estimate $\hat{q}_b = 1$. The resulting decomposition yields a dominant common component that explains approximately 85% of the total variance in the standardized field, indicating that a single shared spatio-temporal factor captures most of the large-scale dependence across variables and pressure levels.

Overall, the results illustrate that GSTFM provides a coherent and interpretable reduction of high-dimensional lattice data while preserving the spatial dependence structure that is lost under naive stacking approaches.

7 Figures

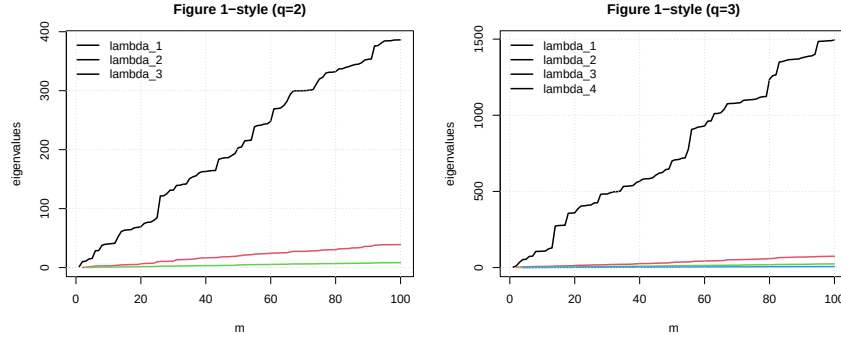


Figure 1: Figure 1 (GSTFM diagnostic). Averaged eigenvalues of the kernel-smoothed spatio-temporal spectral density estimator as the cross-sectional dimension m increases. Under GSTFM, the first q eigenvalues increase strongly while the $(q + 1)$ -th remains relatively stable, producing an eigen-gap that identifies q .

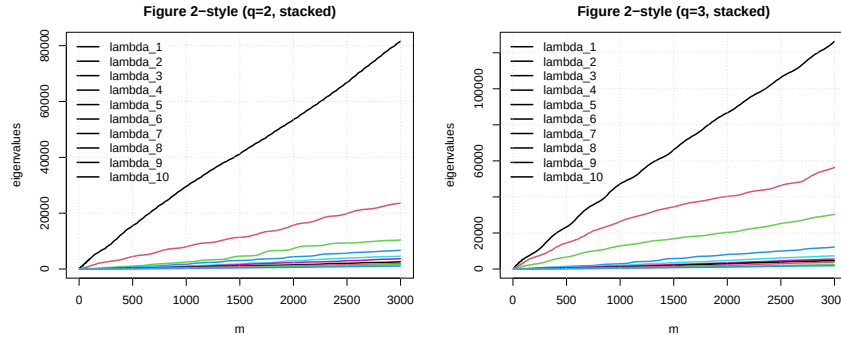


Figure 2: Figure 2 (stacked GDFM diagnostic). Averaged eigenvalues of the spectral density estimator for the stacked time series representation. Unlike GSTFM, the eigen-gap weakens or disappears as the dimension grows, illustrating the loss of identification when spatial structure is ignored.

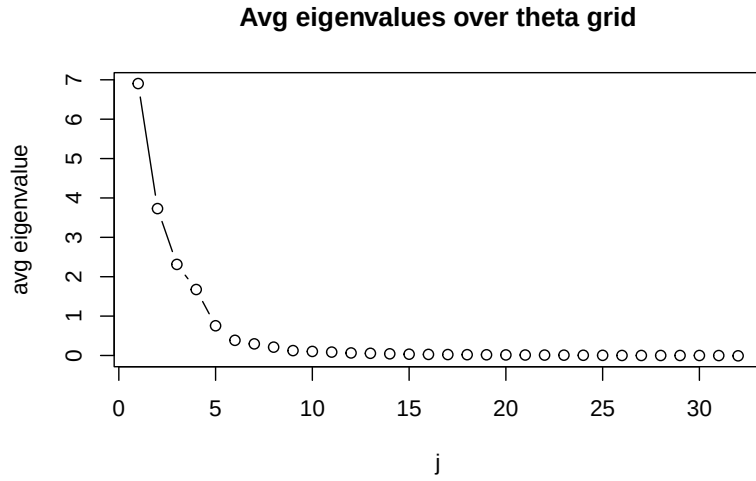


Figure 3: Frequency-averaged eigenvalues of the kernel-smoothed spatio-temporal spectral density estimator. The sharp drop after the first eigenvalue supports the choice $\hat{q} = 1$.

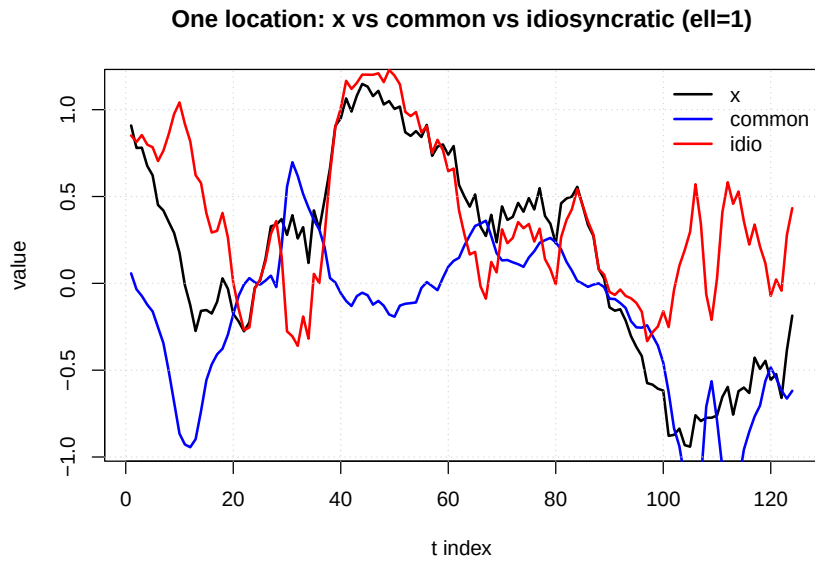


Figure 4: Time-series decomposition at one central spatial location. The observed series (black) is decomposed into the estimated common component (blue) and idiosyncratic component (red).

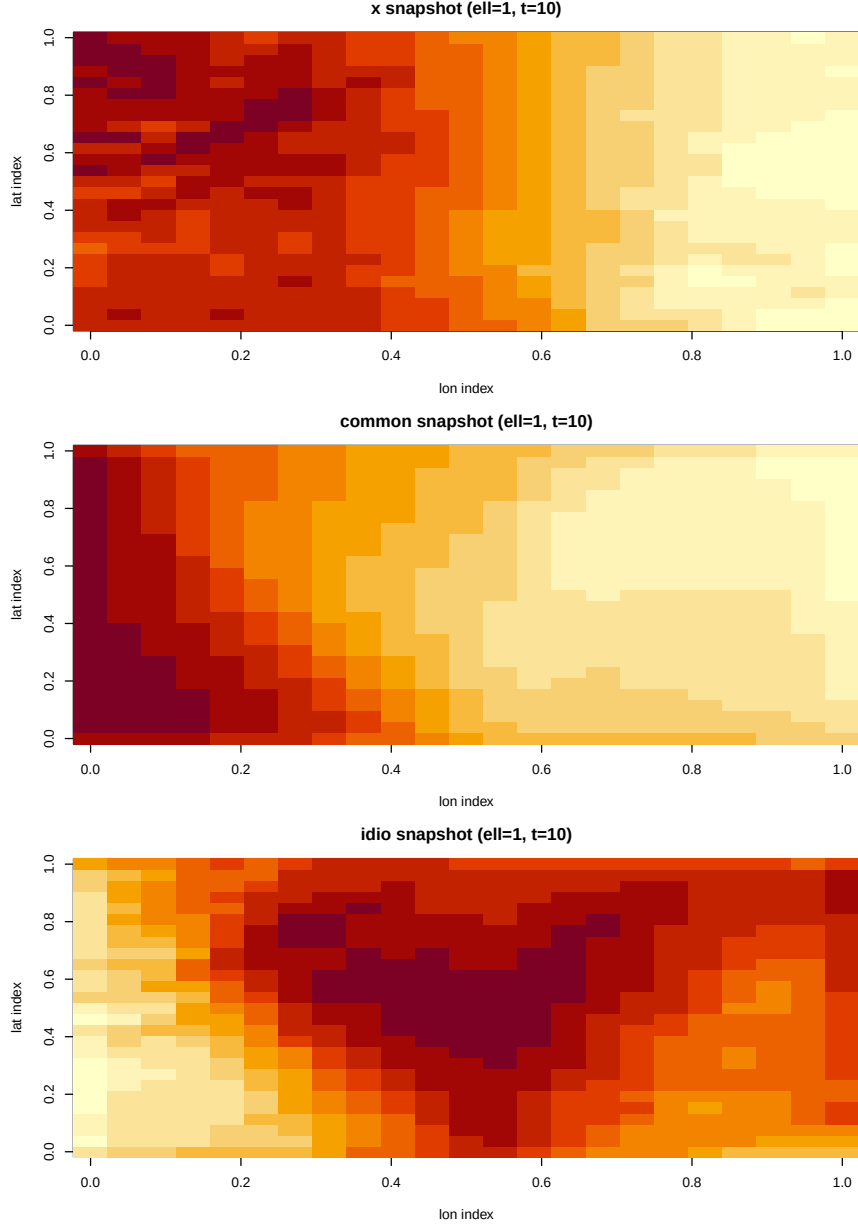


Figure 5: Spatial snapshots at time index $t = 10$ for component $\ell = 1$. Top: standardized field $x^{(sc)}$. Middle: estimated common component $\hat{\chi}$. Bottom: idiosyncratic component $\hat{\xi}$.